OBSTRUCTION RESIDUES AT NEGATIVE-MERSENNE MULTIPLIERS: A SYNCHRONISATION HIERARCHY AND DENSITY BOUNDS

JAKOB STEMBERGER

ABSTRACT. For a Mersenne multiplier $q = 2^{v_M} - 1$ with $v_M \geq 2$ and an odd bias $b \neq 0$, the accelerated map $T_{-q,b}(n) := (-qn + b)/2^{v_2(-qn+b)}$ acts on the odd integers, and the companion $(an + b)$-family manuscript singles it out as the one case admitting a finer structure than the general theory captures. We develop that structure. The starting point is an exact algebraic identity, $-qr + b = 2^v(-qm + b) + (2^{v_M} - 2^v)b$ whenever $r + b = 2^v m$, whose correction term $(2^{v_M} - 2^v)b$ vanishes precisely at $v = v_M$; the vanishing is exclusive to Mersenne multipliers and synchronises a *principal class* of obstructions. Iterating the two-track parallel reduction is governed by a linear recursion on an affine certificate $a_K^{(k)} = \lambda_k a_I^{(k)} + D_k b$, whose synchronising states are parametrised by a signed index $j \in \mathbb{Z}$ as $(\lambda_k, D_k) = (2^{jv_M}, -(2^{jv_M} - 1)/q)$. This yields a sub-class hierarchy indexed by the synchronisation level k and the signed index j: we classify the level-3 sub-classes (three universal families for $v_M \geq 3$, four at $v_M = 2$) with explicit substitution chains, exhibit the level-4 families (including a genuinely signed $j = -1$ family with a rational synchronisation), and prove a node-lift theorem showing that inserting a fixed-point step at the unique fixed point $(\lambda^*, D^*) = (2^{v_M+1}, -1)$ of the recursion lifts each sub-class to the next level, generating four iterated families. Summing the resulting disjoint residue cylinders gives the density lower bound $c_W^{(-q)} \geq 1/(2q)$, which we refine; the same families yield a matching upper bound, making $1/(2q)$ asymptotically sharp as $v_M \to \infty$ — $c_W^{(-q)} = (1/(2q))(1 + O(2^{-v_M}))$ — conditional on a classification-completeness hypothesis. A companion observability bound $|j| \leq (L - k)/v_M$ controls the level at which a sub-class first becomes visible. Each finitely-checkable claim is backed by an executable falsifier — a check that surfaces a counterexample if the claim is false; the limit and asymptotic statements are recorded as such.

Date: May 29, 2026.

2020 *Mathematics Subject Classification.* 11B83, 11S05, 37P05, 37B10, 37B40.

Key words and phrases. Collatz conjecture, $3n \pm 1$ map, $(an + b)$ family, Mersenne multiplier, residue dynamics, parallel reduction, 2-adic density.

1. Introduction

1.1. **Setting.** For odd integers a, b with $|a| \geq 3$ and $b \neq 0$, the *affine accelerated map* $T_{a,b}$ acts on the odd integers by

$$T_{a,b}(n) := \frac{an+b}{2^{v_2(an+b)}}.$$

For $a > 0$ this restricts to a self-map of the positive odd integers $\mathbb{N}_{\mathrm{odd}}$; for $a < 0$ — the case of this paper — an orbit may leave the positives, but the obstruction analysis is purely residue-theoretic (the obstruction sets live in $\{1, \ldots, 2^L\}$, and Section 8 reads their density as a Haar measure on the 2-adic integers $\mathbb{Z}_2$), so the sign of intermediate values is immaterial. Two specialisations are classical: $(a, b) = (3, 1)$ is the Collatz map T_+ [4, 3], and $(a, b) = (3, -1)$ is the $3n - 1$ map T_- [3]. Generalised $(an + b)$ maps go back to Hasse's generalised Syracuse algorithm, whose theory is developed by Matthews and Watts [5] and Möller [6]; the broader class of piecewise-defined generalised Collatz iterations is computationally universal [2], which is part of what makes an exact, level-by-level algebraic characterisation of a structured subfamily worthwhile. The strongest unconditional progress towards the Collatz conjecture is Tao's theorem that almost all orbits attain almost bounded values [10].

This paper studies the *negative-Mersenne multipliers*

$$a = -q, \qquad q := 2^{v_M} - 1 \text{ Mersenne}, \quad v_M \geq 2,$$

so $|a| \in \{3, 7, 15, 31, 63, \ldots\}$. The companion manuscript on the $(an + b)$ family [9] develops a uniform theory of obstruction residues across all odd multipliers with $|a| \geq 3$ — a bias-axis conjugation that makes cardinality and density depend only on a, a multiplier-order lemma, the existence of the asymptotic density, and a no-shift-zero-atom theorem — and explicitly flags the negative-Mersenne case as admitting a *finer* sub-class hierarchy, treated "in a separate manuscript." The present paper is that manuscript. The base case $(a, b) = (3, -1)$, i.e. $v_M = 2$, is the T_--manuscript [8], which proves a constructive lift, positive density, and full factor complexity; here we explain what is special about the whole Mersenne ladder $v_M = 2, 3, 4, \ldots$.

1.2. **Obstruction residues.** We recall the obstruction-residue framework of [9] in the specialisation $a = -q$; a self-contained account is given in Section 2. For an odd residue r and a level L, a two-track parallel reduction runs the map $T_{-q,b}$ simultaneously on r and on its first reduced value m, carrying an affine certificate between the tracks; r is an *obstruction residue* at level L when the certificate vanishes at termination, i.e. when the two tracks synchronise. The set of such residues in $\{1, \ldots, 2^L\}$ is denoted $\mathcal{O}_L^{(-q,b)}$, the *atomic* obstructions (those not lifted from level $L - 1$) are denoted $\mathcal{A}_L^{(-q,b)}$, and the asymptotic density is

$$c_W^{(-q)} := \lim_{L\to\infty} \frac{|\mathcal{O}_L^{(-q,b)}|}{2^L},$$

which exists by [9]. The fine invariant of an obstruction is its *synchronisation level* k — the index at which the two tracks first agree — together with a *signed synchronisation index* $j \in \mathbb{Z}$ recording the cumulative valuation gap between the tracks.

1.3. **Main results.** The Mersenne condition enters through a single exact identity (Theorem 3.1): for $r + b = 2^v m$,

$$-qr + b = 2^v(-qm + b) + (2^{v_M} - 2^v)\, b,$$

whose correction term $(2^{v_M} - 2^v)b$ vanishes *exactly* at $v = v_M$, and only for Mersenne q does such a v exist for arbitrary b (Corollaries 3.2 and 3.3). The vanishing produces the *principal synchronising class* — the residues with $v_2(r + b) = v_M$, which synchronise in a single step; iterating it is governed by a linear recursion on the affine certificate (Theorem 4.1), whose synchronising states are exactly $(\lambda_k, D_k) = (2^{jv_M}, -(2^{jv_M} - 1)/q)$, $j \in \mathbb{Z}$.

This recursion structures the obstructions into a sub-class hierarchy:

- a level-3 structure theorem (Theorem 5.1): for $v_M \geq 3$ exactly three universal families A, B, C, with explicit substitution chains, and a fourth family at $v_M = 2$;
- the level-4 families (Theorem 6.1), including a genuinely signed family G_1 with $j = -1$ that synchronises rationally;
- a node-lift theorem (Theorem 7.1): the recursion has a unique fixed point $(\lambda^*, D^*) = (2^{v_M+1}, -1)$, and inserting the fixed-point step at a node lifts a sub-class to the next level, generating four iterated families $A_k, B_k, C_k, F_{1,k}$;
- a density lower bound $c_W^{(-q)} \geq 1/(2q)$ (Theorem 8.2), obtained by summing disjoint residue cylinders, refined in Corollary 8.3 and shown asymptotically sharp under a classification-completeness hypothesis (Theorem 10.1): $c_W^{(-q)} = (1/(2q))(1 + O(2^{-v_M}))$;
- an observability bound $|j| \leq (L - k)/v_M$ (Theorem 9.1) controlling the level at which a level-k, index-j sub-class first becomes visible.

The density scale deserves emphasis. With $|a| = 2^{v_M} - 1$ the multiplicative order $e := \mathrm{ord}_{|a|}(2)$ equals v_M, so the bound $1/(2q) = \Theta(2^{-v_M}) = \Theta(2^{-e})$ is the *square root* of the empirical $\Theta(4^{-e})$ density observed at generic positive multipliers in [9]: negative-Mersenne multipliers are anomalously obstruction-rich, by an exponential factor, and the present hierarchy is the structural reason.

1.4. **Relation to the family manuscript.** We use the obstruction-residue framework, the parallel reduction, and the existence of the asymptotic density from [9] as a black box (recalled in Section 2); everything specific to the Mersenne ladder — the main identity, the recursion, the sub-class families, the node lift, and the density bounds — is proved here. The family manuscript's no-shift-zero-atom theorem holds "outside the negative-Mersenne case": its proof forces a shift-zero obstruction to take $J \geq 2$ steps unless $2^v = |a| + 1$ is solvable, which happens exactly for Mersenne $|a|$ (at $v = v_M$). That one-step exception is the principal class of the present paper, so the family theorem excludes precisely the negative-Mersenne ladder and the two accounts are complementary rather than overlapping.

1.5. **Outline.** Section 2 fixes the parallel reduction and the affine certificate. Section 3 proves the main identity and its Mersenne/non-Mersenne corollaries. Section 4 derives the recursion theorem and identifies its synchronising states. Sections 5 and 6 classify the level-3 and level-4 sub-classes. Section 7 proves the node-lift theorem. Section 8 establishes the density lower bound and its refinement, Section 9 the observability bound, and Section 10 the asymptotic sharpness. Section 11 collects open problems.

2. Setup: parallel reduction and the affine certificate

Fix throughout a Mersenne multiplier $q = 2^{v_M} - 1$ with $v_M \geq 2$ and an odd bias $b \neq 0$; we write $a = -q$ and study $T_{-q,b}$. The family-existence results and the density lower bounds below hold for any such b; the *converse* classification of synchronising states (Theorem 4.1) and the exact sub-class counts it yields (Theorem 5.1) assume in addition that b is coprime to q — satisfied by the classical biases $b = \pm 1$ and by every bias used in the verification.

2.1. **Nesting and the two-track reduction.** Let r be a positive odd integer. Since b is odd, $r + b$ is even; write

$$v_0 := v_2(r+b) \geq 1, \qquad m_0 := (r+b)/2^{v_0} \text{ (odd)}.$$

The *parallel reduction* at level $L > v_0$ runs two tracks. The *K-track* carries the value $a_K^{(0)} := r$ and modulus exponent L; the *I-track* carries $a_I^{(0)} := m_0$ and modulus exponent $L - v_0$. At each step $k \geq 0$ set

$$v_K^{(k)} := v_2\big(-q\, a_K^{(k)} + b\big), \qquad v_I^{(k)} := v_2\big(-q\, a_I^{(k)} + b\big),$$

and update

$$a_K^{(k+1)} := \frac{-q\, a_K^{(k)} + b}{2^{v_K^{(k)}}}, \qquad a_I^{(k+1)} := \frac{-q\, a_I^{(k)} + b}{2^{v_I^{(k)}}},$$

each step being admissible while the corresponding modulus exponent stays positive. The cumulative valuations are $V_K^{(k)} := \sum_{i<k} v_K^{(i)}$ and $V_I^{(k)} := \sum_{i<k} v_I^{(i)}$, so the residual modulus exponents are $L - V_K^{(k)}$ and $L - v_0 - V_I^{(k)}$.

2.2. **The affine certificate.**

Lemma 2.1 (Affine certificate)**.** *For every step $k \geq 0$ there are $\lambda_k \in \mathbb{Q}^\times$ and $D_k \in \mathbb{Q}$ with*

$$a_K^{(k)} = \lambda_k\, a_I^{(k)} + D_k\, b \qquad \text{in } \mathbb{Q},$$

and $\lambda_0 = 2^{v_0}$, $D_0 = -1$.

Proof. From $r + b = 2^{v_0} m_0$ we get $r = 2^{v_0} m_0 - b$, i.e. $a_K^{(0)} = 2^{v_0} a_I^{(0)} - b$, the claim at $k = 0$ with $(\lambda_0, D_0) = (2^{v_0}, -1)$. For the inductive step, substitute the affine form $a_K^{(k)} = \lambda_k a_I^{(k)} + D_k b$ into $-q\, a_K^{(k)} + b$ and divide by $2^{v_K^{(k)}}$: this rewrites $a_K^{(k+1)}$ once more as a $\mathbb{Q}$-affine function of $a_I^{(k+1)}$ and b, so the form persists with some $(\lambda_{k+1}, D_{k+1}) \in \mathbb{Q}^\times \times \mathbb{Q}$. The explicit update for (λ_{k+1}, D_{k+1}) is computed in Theorem 4.1. □

Definition 2.2 (Synchronisation level and signed index)**.** The reduction *synchronises* at the least index k with $a_K^{(k)} = a_I^{(k)}$, equivalently $(\lambda_k, D_k) = (1, 0)$; this k is the *synchronisation level*. The *signed synchronisation index* is the unique integer j for which the state one step before synchronisation has $\lambda_{k-1} = 2^{j v_M}$; this is well-defined because Theorem 4.1 shows every pre-synchronisation λ is the integral power $2^{j v_M}$, $j \in \mathbb{Z}$. A residue r is an *obstruction residue* at level L, $r \in \mathcal{O}_L^{(-q,b)}$, when the two tracks synchronise before either modulus exponent is exhausted; $\mathcal{A}_L^{(-q,b)} \subseteq \mathcal{O}_L^{(-q,b)}$ are those not arising as a lift of a level-$(L-1)$ obstruction, and

$$c_W^{(-q)} := \lim_{L\to\infty} |\mathcal{O}_L^{(-q,b)}|/2^L.$$

Remark 2.3. Definition 2.2 specialises the obstruction residues of [9] to $a = -q$: there the certificate is written $a_I = c\, a_K + d$ and the vanishing condition $\Phi_J = 0$ is the synchronisation $a_K = a_I$ here. The family's shift index δ (the valuation gap at termination) and the signed synchronisation index j used here (read off one step before synchronisation) are different invariants. What the family's no-shift-zero-atom theorem excludes *outside* the negative-Mersenne case is a one-step ($J = 1$) shift-zero obstruction: its proof forces $J \geq 2$ unless $2^v = |a| + 1$ is solvable, which holds exactly for Mersenne $|a|$ at $v = v_M$. That solvable case is the principal Mersenne-synchronising class here — which carries $j = 1$, not $j = 0$. The Mersenne ladder populates $j \in \mathbb{Z}$ (the level-3 families have $j \in \{1, 2\}$, the level-4 family G_1 has $j = -1$), so the signedness of j is essential. Existence of the limit $c_W^{(-q)}$ is the family's density-limit proposition, used as a black box.

Definition 2.4 (Universal vs. q-specific sub-classes)**.** A *sub-class* is a set of obstruction residues sharing one initial nesting exponent v_0 and one driving sequence of valuation pairs $(v_K^{(i)}, v_I^{(i)})$. The sub-class is *universal* when this valuation data is *affine in* v_M — each of $v_0, v_K^{(i)}, v_I^{(i)}$ is an integer-affine function $\alpha v_M + \beta$ with fixed integers $\alpha \geq 0$ and β independent of v_M (so a constant β, the offset form $v_M + \beta$, or e.g. $2v_M$ are all admitted) — so that a single closed form describes it for all v_M in its range; otherwise it is *q-specific*. The exhaustiveness statements below ("exactly three families", etc.) are understood *within the universal sub-classes*: they enumerate the affine-in-v_M valuation patterns and do not assert an a-priori bound ruling out non-affine families. The number of q-specific sub-classes is itself v_M-dependent (Theorem 6.1, Conjecture 11.1).

3. The main identity

Theorem 3.1 (Main identity). *Let $q = 2^{v_M} - 1$. For all integers r, m, b and all $v \geq 1$ with $r + b = 2^v m$,*

$$-qr + b = 2^v(-qm + b) + \left(2^{v_M} - 2^v\right) b.$$

Proof. From $r = 2^v m - b$,

$$-qr + b = -q\left(2^v m - b\right) + b = -q\,2^v m + (q+1)\,b.$$

Since $q + 1 = 2^{v_M}$,

$$-q\,2^v m + 2^{v_M} b = 2^v(-qm) + 2^v b + \left(2^{v_M} - 2^v\right) b = 2^v(-qm + b) + \left(2^{v_M} - 2^v\right) b.$$

□

Corollary 3.2 (Mersenne synchronisation). *At $v = v_M$ the correction term vanishes and $-qr + b = 2^{v_M}(-qm + b)$. Moreover, among all $q \geq 2$, a level v with $(q + 1 - 2^v)\,b = 0$ for arbitrary b exists if and only if $q = 2^{v_M} - 1$ is Mersenne, in which case $v = v_M$ is the unique such level.*

Proof. Setting $v = v_M$ gives $2^{v_M} - 2^v = 0$, hence the displayed identity. For the characterisation: $(q+1-2^v)b = 0$ for $b \neq 0$ forces $q+1 = 2^v$, i.e. $q+1$ is a power of two; this holds for some v iff q is Mersenne, and then, by uniqueness of the binary representation, $v = v_M = v_2(q+1)$ is the only such level. □

Corollary 3.3 (Non-Mersenne asymmetry). *For an arbitrary integer $q \geq 2$ and $r + b = 2^v m$,*

$$-qr + b = 2^v(-qm + b) + (q + 1 - 2^v)\,b.$$

If q is not Mersenne, then $q+1-2^v \neq 0$ for every $v \geq 1$, so for every odd $b \neq 0$ the correction term is nonzero at all levels and no principal synchronising class exists.

Proof. The displayed identity is the computation in the proof of Theorem 3.1 with $q + 1$ in place of 2^{v_M}. If q is not Mersenne then $q + 1$ is not a power of two, so $q + 1 \neq 2^v$ for all v; thus $(q + 1 - 2^v)b \neq 0$ for odd $b \neq 0$. □

The finite arithmetic of Theorem 3.1 and Corollaries 3.2–3.3 is exercised over many samples by `code/python/identity.py`.

4. The recursion theorem

Theorem 4.1 (Recursion theorem). *Let $q = 2^{v_M} - 1$, and suppose $a_K^{(k)} = \lambda_k a_I^{(k)} + D_k b$. (The converse classification of synchronising states below assumes in addition $\gcd(b, q) = 1$; cf. Section 2.) Then*

$$\lambda_{k+1} = \lambda_k \cdot 2^{v_I^{(k)} - v_K^{(k)}}, \qquad D_{k+1} = \frac{1 - \lambda_k - q\,D_k}{2^{v_K^{(k)}}},$$

with initial data $(\lambda_0, D_0) = (2^{v_0}, -1)$. Integrality of D_{k+1} is the consistency condition $2^{v_K^{(k)}} \mid E_k$ with $E_k := 1 - \lambda_k - qD_k$. The reduction synchronises

($a_K^{(k)} = a_I^{(k)}$) iff $(\lambda_k, D_k) = (1, 0)$*, and the states one step before synchronisation are, assuming* $\gcd(b, q) = 1$*, exactly*

$$(\lambda_k, D_k) = \Big(2^{jv_M},\ -\tfrac{2^{jv_M}-1}{q}\Big), \qquad j \in \mathbb{Z},$$

each of which is integral because $q = 2^{v_M} - 1 \mid 2^{jv_M} - 1$.

Proof. Substituting $a_K^{(k)} = \lambda_k a_I^{(k)} + D_k b$ into $-qa_K^{(k)} + b$ and using $q+1 = 2^{v_M}$ only at the end,

$$-q\,a_K^{(k)} + b = \lambda_k\big(-q\,a_I^{(k)} + b\big) + \big(1 - \lambda_k - qD_k\big)b = \lambda_k\big(-q\,a_I^{(k)} + b\big) + E_k\,b.$$

By definition of the step, $-q\,a_K^{(k)} + b = 2^{v_K^{(k)}} a_K^{(k+1)}$ and $-q\,a_I^{(k)} + b = 2^{v_I^{(k)}} a_I^{(k+1)}$; hence

$$2^{v_K^{(k)}} a_K^{(k+1)} = \lambda_k\, 2^{v_I^{(k)}} a_I^{(k+1)} + E_k\, b,$$

and dividing by $2^{v_K^{(k)}}$ gives the stated updates with $\lambda_{k+1} = \lambda_k 2^{v_I^{(k)} - v_K^{(k)}}$ and $D_{k+1} = E_k / 2^{v_K^{(k)}}$. Since $a_K^{(k+1)}, a_I^{(k+1)} \in \mathbb{Z}$, the quantity D_{k+1} is an integer iff $2^{v_K^{(k)}} \mid E_k$. The initial data are Lemma 2.1.

Synchronisation $a_K^{(k)} = a_I^{(k)}$ for all admissible $a_I^{(k)}$ is $\lambda_k = 1$, $D_k = 0$. Now take a state $(\lambda_k, D_k) = (2^{jv_M}, -(2^{jv_M} - 1)/q)$. Then

$$E_k = 1 - 2^{jv_M} - q\Big(-\tfrac{2^{jv_M}-1}{q}\Big) = 1 - 2^{jv_M} + (2^{jv_M} - 1) = 0,$$

so $D_{k+1} = 0$; choosing the step with $v_I^{(k)} - v_K^{(k)} = -jv_M$ gives $\lambda_{k+1} = 2^{jv_M} \cdot 2^{-jv_M} = 1$, i.e. the next step synchronises.

Conversely — here we use $\gcd(b, q) = 1$ (the standing coprimality assumption of Section 2) — suppose the next step synchronises: $\lambda_{k+1} = 1$ and $D_{k+1} = 0$, with common synchronised value $w := a_K^{(k+1)} = a_I^{(k+1)}$. From $\lambda_{k+1} = \lambda_k 2^{v_I^{(k)} - v_K^{(k)}} = 1$ we get $\lambda_k = 2^s$ with $s := v_K^{(k)} - v_I^{(k)} \in \mathbb{Z}$, and $E_k = 0$ gives $D_k = (1 - 2^s)/q$. We show $v_M \mid s$ *for either sign of* s, so that $j := s/v_M \in \mathbb{Z}$. Inverting the synchronising step,

$$a_K^{(k)} = \frac{b - 2^{v_K^{(k)}} w}{q}, \qquad a_I^{(k)} = \frac{b - 2^{v_I^{(k)}} w}{q},$$

are both integers, so q divides their difference $(2^{v_I^{(k)}} - 2^{v_K^{(k)}})\,w$. Writing $m := \min(v_K^{(k)}, v_I^{(k)})$, this difference equals $\pm\,2^m(2^{|s|} - 1)\,w$, and w is itself coprime to q — a common prime factor of w and q would divide $q\,a_K^{(k)} = b - 2^{v_K^{(k)}} w$, hence b, contradicting $\gcd(b, q) = 1$. Since q is odd, it follows that $q \mid 2^{|s|} - 1$. Now 2 is a unit of multiplicative order exactly v_M modulo q: indeed $2^{v_M} = q + 1 \equiv 1 \pmod{q}$, while $0 < 2^t - 1 < 2^{v_M} - 1 = q$ for $0 < t < v_M$, so that $2^t \equiv 1 \pmod{q}$ holds if and only if $v_M \mid t$, for every integer $t \geq 0$. Applying this with $t = |s|$ gives $v_M \mid s$, whence $\lambda_k = 2^{jv_M}$ and $D_k = -(2^{jv_M} - 1)/q$ with $j \in \mathbb{Z}$ — the negative-j states being the rational ones ($\lambda_k = 2^{jv_M} < 1$) realised by the G_1 family of Section 6. □

The recursion and its consistency arithmetic are checked over many step sequences by `code/python/recursion.py`.

5. The level-3 structure theorem

Theorem 5.1 (Level-3 structure). *Let $q = 2^{v_M} - 1$ with $\gcd(b, q) = 1$ (Section 2), so that the converse of Theorem 4.1 applies. For $v_M \geq 3$ the obstruction residues that synchronise at level $k = 3$ fall into exactly three universal families, with the data*

family	v_0	$\big(v_K^{(0)}, v_I^{(0)}\big)$, $\big(v_K^{(1)}, v_I^{(1)}\big)$	j	(λ_2, D_2)
A	$v_M + 1$	(v_M, v_M), $(v_M, v_M - 1)$	1	$(2^{v_M}, -1)$
B	$v_M - 1$	$(v_M - 1, 1)$, $(v_M, 2v_M - 1)$	1	$(2^{v_M}, -1)$
C	1	$(1, v_M + 1)$, $(v_M - 1, 2v_M - 2)$	2	$(2^{2v_M}, -(2^{v_M} + 1))$

and synchronisation identity $a_K^{(2)} - 2^{jv_M} a_I^{(2)} = -\frac{2^{jv_M-1}}{q}\, b$ (equal to $-b$ for A, B and $-(2^{v_M} + 1)\, b$ for C). The residues of families A and B are the arithmetic progressions

$$A: \quad r = 2^{3v_M}\nu - K_A\, b, \quad K_A = 2^{2v_M+2} + 2^{v_M+1} + 1, \qquad \nu \text{ odd};$$

$$B: \quad r = 2^{3v_M}\tau - K_B\, b, \quad K_B = 5 \cdot 2^{2v_M-1} + 3 \cdot 2^{v_M-1} + 1, \quad \tau \in \mathbb{Z}.$$

The constraint ν odd in family A selects its atomic residues — those whose leading nesting coefficient is odd, halving the cylinder and giving the per-level density $2^{-(kv_M+1)}$ — whereas family B's cylinder is full ($\tau \in \mathbb{Z}$, per-level density 2^{-kv_M}); this asymmetry is the source of the differing family contributions to the density bound of Section 8. For $v_M = 2$ there is a fourth family D with $v_0 = v_M + 3 = 5$ and $j = 3$.

Proof. A level-3 sub-class (Definition 2.4) is a choice of initial nesting exponent v_0 and a sequence of valuation pairs $(v_K^{(0)}, v_I^{(0)}), (v_K^{(1)}, v_I^{(1)})$ driving the recursion of Theorem 4.1 from $(\lambda_0, D_0) = (2^{v_0}, -1)$ to a state (λ_2, D_2) one step before synchronisation, hence of the form $(2^{jv_M}, -(2^{jv_M} - 1)/q)$. Running the recursion symbolically in v_M for each of the three displayed (v_0, pair-sequence) data yields exactly the listed (λ_2, D_2), and back-substituting through the nesting recovers the stated arithmetic progressions; the synchronisation identity is the relation $a_K^{(2)} = \lambda_2 a_I^{(2)} + D_2 b$ read off from (λ_2, D_2).

For completeness we enumerate the admissible *universal* valuation data (Definition 2.4, affine in v_M; the families above realise the offsets $0, \pm 1, \pm 2$). By Theorem 4.1 the two steps must drive $(2^{v_0}, -1)$ to a pre-synchronisation state $(2^{jv_M}, -(2^{jv_M} - 1)/q)$, and by the observability bound (Theorem 9.1) a level-3 sub-class visible at any level has $|j|$ bounded; the consistency relations $2^{v_K^{(i)}} \mid E_i$ then leave finitely many affine offset patterns to test, each a polynomial identity in v_M. Enumerating them — symbolically in v_M (which yields exactly A, B, C for $v_M \geq 3$) and concretely for $q = 3, 7$ via `code/python/stage3.py` — shows there are no others. At $v_M = 2$ the families B ($v_0 = v_M - 1 = 1$) and C ($v_0 = 1$) share the same initial nesting exponent — the offset $v_M - 2$ separating their nesting depths collapses to 0 — and an additional valuation pattern with $v_0 = 5$ becomes admissible, giving the extra family D ($j = 3$); the concrete counts confirm this — four sub-classes summing to 132 obstruction residues at $L = 12$ for $q = 3$, three summing to 64 at $L = 14$ for $q = 7$. The general-v_M exhaustiveness rests on this affine-shape enumeration rather than on an a-priori bound ruling out non-affine families. □

6. The level-4 families

Theorem 6.1 (Level-4 families)**.** *Let $q = 2^{v_M} - 1$ with $v_M \geq 3$. There are at least five universal families synchronising at level $k = 4$:*

family	v_0	$(v_K^{(0)}, v_I^{(0)})$, $(v_K^{(1)}, v_I^{(1)})$, $(v_K^{(2)}, v_I^{(2)})$	j
A_{lift}	$v_M + 1$	(v_M, v_M), (v_M, v_M), $(v_M, v_M - 1)$	1
B_{lift}	$v_M - 1$	$(v_M - 1, 1)$, $(v_M, 2v_M)$, $(v_M, v_M - 1)$	1
C_{lift}	1	$(1, v_M + 1)$, $(v_M - 1, 2v_M - 1)$, $(2v_M, v_M - 1)$	1
F_1	$v_M + 1$	$(v_M, v_M - 2)$, $(v_M - 1, 1)$, $(v_M, 2v_M - 1)$	1
G_1	$v_M - 1$	$(v_M - 1, 1)$, $(v_M + 1, v_M - 1)$, $(2v_M - 1, v_M)$	-1

The four $j = 1$ families satisfy $a_K^{(3)} - 2^{v_M} a_I^{(3)} = -b$, while G_1 ($j = -1$) satisfies the rational synchronisation $a_K^{(3)} - 2^{-v_M} a_I^{(3)} = 2^{-v_M} b$. In addition there is a v_M-dependent number of further q-specific level-4 sub-classes — six at $v_M = 3$, four at $v_M = 4$ — which are not universal in v_M; their count is not uniform in v_M.

Proof. For each of the five displayed exponent sequences, running the recursion of Theorem 4.1 symbolically in v_M from $(\lambda_0, D_0) = (2^{v_0}, -1)$ produces a pre-synchronisation state $(\lambda_3, D_3) = (2^{jv_M}, -(2^{jv_M} - 1)/q)$ with the listed j; for $j = 1$ this is $(2^{v_M}, -1)$, giving $a_K^{(3)} - 2^{v_M} a_I^{(3)} = -b$, and for G_1 the value $j = -1$ gives $\lambda_3 = 2^{-v_M}$ and $D_3 = -(2^{-v_M} - 1)/q = 2^{-v_M}$, the displayed rational synchronisation — the first sub-class requiring a negative signed index, which is why the index must be taken in $\mathbb{Z}$ rather than $\mathbb{N}$. The q-specific families are found by the same recursion for fixed small v_M but do not admit a closed form uniform in v_M, and their number varies with v_M (six at $v_M = 3$, four at $v_M = 4$). The exact count of q-specific families is not used in any later bound — the density results of Section 8 only require the listed families to be present and pairwise disjoint, so the lower bounds hold regardless of how many further families exist. The symbolic sync identities of the five universal families are verified by `code/python/recursion.py`; the exhaustive level-4 sub-class count — 11 at $v_M = 3$ and 9 at $v_M = 4$, whence six and four q-specific families respectively — is verified by `code/python/stage4_enum.py`. □

7. The node-lift theorem

Theorem 7.1 (Node lift)**.** *Let $q = 2^{v_M} - 1$. The recursion of Theorem 4.1 has a unique fixed point carrying the synchronisation constant $-b$ (i.e. with $D = -1$),*

$$(\lambda^*, D^*) = (2^{v_M+1}, -1),$$

attained by the step $(v_K, v_I) = (v_M, v_M)$. If a level-k sub-class passes through this A-node at some position ℓ, then inserting one further fixed-point step (v_M, v_M) at position ℓ produces a level-$(k+1)$ sub-class with the same synchronisation constant $-b$. Consequently there are four iterated families

$$A_k\ (k \geq 2), \quad B_k\ (k \geq 4), \quad C_k\ (k \geq 5), \quad F_{1,k}\ (k \geq 4),$$

each satisfying $a_K^{(k-1)} - 2^{v_M} a_I^{(k-1)} = -b$, generated by iterating the A-node from the universal families of Sections 5–6 that thread it.

Proof. At the state $(\lambda, D) = (2^{v_M+1}, -1)$ a step with $(v_K, v_I) = (v_M, v_M)$ has $v_I - v_K = 0$, so $\lambda' = \lambda$, and

$$E = 1 - \lambda - qD = 1 - 2^{v_M+1} + q = 1 - 2^{v_M+1} + (2^{v_M} - 1) = -2^{v_M},$$

whence $D' = E/2^{v_K} = -2^{v_M}/2^{v_M} = -1$. Thus $(\lambda', D') = (2^{v_M+1}, -1) = (\lambda, D)$: the A-node is fixed. It is moreover the unique fixed point carrying the synchronisation constant $-b$, i.e. with $D = -1$. Indeed, solving $\lambda' = \lambda$, $D' = D$ for such a state: $\lambda' = \lambda$ forces $v_I = v_K$ (as $\lambda' = \lambda 2^{v_I - v_K}$ with $\lambda \neq 0$), and then $D' = D = -1$ reads $E = 1 - \lambda + q = -2^{v_K}$, i.e.

$$\lambda = 1 + q + 2^{v_K} = 2^{v_M} + 2^{v_K}$$

(using $1 + q = 2^{v_M}$). A sum of two powers of two is itself a power of two if and only if the two exponents are equal, so $v_K = v_M$ and $\lambda = 2^{v_M+1}$; the A-node is the only such state, with no restriction on the valuation range required.

Because the inserted step neither changes (λ, D) nor the eventual synchronising value $(2^{v_M}, -1)$ reached one step later, a level-k sub-class threading the node at position ℓ lifts to a level-$(k+1)$ sub-class with synchronisation constant $-b$ unchanged. Four universal families thread the A-node: A_2 and $F_{1,4}$ start *at* it (their $v_0 = v_M + 1$ gives $(\lambda_0, D_0) = (2^{v_M+1}, -1)$), while B_4 and C_5 reach it after some steps. Iterating the fixed-point insertion from each generates the towers A_k $(k \geq 2)$, $F_{1,k}$ $(k \geq 4)$, B_k $(k \geq 4)$, C_k $(k \geq 5)$ at the stated minimal levels, all with synchronisation constant $-b$. We do not claim these are the only node-threading sub-classes at every level: the lower bounds of Section 8 use only their presence and pairwise disjointness, and the asymptotic upper bound is conditional on the completeness hypothesis of Theorem 10.1. The fixed-point computation and the iterated synchronisation identities are verified symbolically in v_M up to $k = 8$ by `code/python/recursion.py`. □

Remark 7.2. The level-4 family C_{lift} of Theorem 6.1 (valuation sequence ending $(v_K^{(2)}, v_I^{(2)}) = (2v_M, v_M - 1)$) and the base C_5 of the C-tower (ending

$(2v_M, v_M)$, synchronising at level 5) are distinct families that happen to share the letter C; only C_5 threads the A-node.

8. Density lower bound

Remark 8.1 (2-adic reading of the density). By the lift theorem of [9] the obstruction sets are lift-stable, so they define an open subset of the 2-adic integers $\mathbb{Z}_2$ (an increasing union of residue cylinders, the density being a non-decreasing limit), and $c_W^{(-q)}$ is its normalised Haar measure — the negative-Mersenne instance of the 2-adic density placed in the Bernstein–Lagarias tradition [1] (for the 2-adic background see [7]). The bounds below are therefore measure lower bounds on a fixed subset of $\mathbb{Z}_2$.

Theorem 8.2 (Density lower bound)**.** *Let $q = 2^{v_M} - 1$ with $v_M \geq 2$. Then*

$$c_W^{(-q)} \geq \frac{1}{2q}.$$

Proof. By Corollary 3.2 the principal Mersenne-synchronising class is exactly the residues with $v_2(r+b) = v_M$ (those synchronise in a single step), a cylinder of density $2^{-v_M} - 2^{-v_M-1} = 2^{-v_M-1}$, and by Theorem 7.1 the iterated family A_k $(k \geq 2)$ occupies, for each k, a residue cylinder of density 2^{-kv_M-1}; identifying the principal class as the $k = 1$ term of the same geometric pattern, these cylinders are pairwise disjoint, because the synchronisation level is by definition the *least* index at which the two tracks agree and is thus a well-defined function of r, so each obstruction lies in the cylinder of exactly one k. Summing,

$$c_W^{(-q)} \geq \sum_{k=1}^{\infty} 2^{-kv_M-1} = \frac{2^{-v_M-1}}{1-2^{-v_M}} = \frac{2^{-v_M-1}\,2^{v_M}}{2^{v_M}-1} = \frac{1}{2(2^{v_M}-1)} = \frac{1}{2q}. \qquad \square$$

Corollary 8.3 (Refined lower bound)**.** *Let $v_M \geq 3$. Counting, beyond the principal class and the A-family, the full B-, C- and F-families (each the geometric tower of its members across synchronisation levels) and the family G_1,*

$$c_W^{(-q)} \geq \frac{1}{2q} + \frac{2^{-2v_M}}{q} + \frac{2^{-2v_M-1}}{q} + \frac{2^{-3v_M}}{q} + 2^{-4v_M},$$

the four correction terms being the totals of the B-, C-, F- and G_1-families respectively.

Proof. Each universal family is a tower of residue cylinders, one per synchronisation level k at and above its minimal level, and each cylinder's density is read off from its modulus $2^{\max(V_K^{(k-1)},\, v_0+V_I^{(k-1)})+1}$. By synchronisation level the towers are pairwise disjoint, and within a level distinct families have distinct valuation data (the nesting v_0 *together with* the valuation sequence — v_0 alone does not separate them, e.g. A_{lift} and F_1 share $v_0 = v_M + 1$), so all the cylinders below are disjoint. The members of each family letter

v_M	q	$1/(2q)$	refined bound	$\lvert\mathcal{O}_L^{(-q,b)}\rvert/2^L$, $L=16$
2	3	0.1667	0.2070	0.2498
3	7	0.0714	0.0753	0.0770
4	15	0.0333	0.0338	0.0338
5	31	0.0161	0.0162	0.0162

TABLE 1. The simple bound $1/(2q)$, the refined bound of Corollary 8.3, and the finite density $\lvert\mathcal{O}_L^{(-q,b)}\rvert/2^L$ at $L=16$ (with $b=1$). Both bounds are lower bounds on the limit $c_W^{(-q)}$; the finite density is itself a monotone lower bound (Theorem 8.2), still increasing for small v_M and essentially converged for $v_M \geq 4$, where it agrees with the refined bound to the displayed precision. (For large v_M the finite density at a fixed level can sit just below the refined bound, since both approach $c_W^{(-q)}$ from below.)

have a fixed per-level density, giving a geometric series with ratio 2^{-v_M} and $\sum_{k\geq k_0} 2^{-kv_M} = 2^{-k_0 v_M}\, 2^{v_M}/q$:

family	min. level k_0	per-level density	total
B	3	2^{-kv_M}	$\sum_{k\geq 3} 2^{-kv_M} = 2^{-2v_M}/q$
C	3	$2^{-(kv_M+1)}$	$\sum_{k\geq 3} 2^{-(kv_M+1)} = 2^{-2v_M-1}/q$
F	4	2^{-kv_M}	$\sum_{k\geq 4} 2^{-kv_M} = 2^{-3v_M}/q$
G_1	4	2^{-4v_M}	2^{-4v_M}

(the A-family, levels $k \geq 1$ with per-level density $2^{-(kv_M+1)}$, already sums to $1/(2q)$ in Theorem 8.2). Adding the four totals to $1/(2q)$ gives the stated bound. The level-3 members of the B- and C-families are the families B, C of Theorem 5.1; the level-4 members include B_{lift}, C_{lift}, F_1, G_1 of Theorem 6.1; the higher members are the node-lift towers of Theorem 7.1. The per-family densities and the identity of the total with the displayed bound are checked by `code/python/refined_decomposition.py`, and the table values by `code/python/atom_density.py`. (At $v_M = 2$ the tower decomposition degenerates — families B and C then share $v_0 = 1$ — so the bound is stated for $v_M \geq 3$; the $v_M = 2$ entry of Table 1 is the formula evaluated at $v_M = 2$, which remains a lower bound on $c_W^{(-q)}$ because the monotone finite density of Theorem 8.2 already exceeds it.) □

Table 1 compares both bounds with the finite density; the values are recomputed and checked against the table by `code/python/atom_density.py`.

9. The observability bound

Theorem 9.1 (Observability bound). *Let r be a residue in a level-k sub-class with signed synchronisation index j. Then r has modulus at least $2^{\,k+|j|\,v_M}$, so it contributes to $\mathcal{O}_L^{(-q,b)}$ only for*

$$L \ge k + |j|\,v_M, \qquad \textit{equivalently} \qquad |j| \le \frac{L-k}{v_M}.$$

Proof. Write $A := V_K^{(k-1)}$ and $B := v_0 + V_I^{(k-1)}$ for the cumulative modulus consumption of the two tracks up to synchronisation. The signed index satisfies $jv_M = B - A$, so $|A-B| = |j|\,v_M$; since synchronisation at level k takes $k-1$ steps and each consumes at least one unit of valuation on each track — the iterates $a_K^{(i)}, a_I^{(i)}$ are odd and b is odd, so $-q\,a_K^{(i)} + b$ and $-q\,a_I^{(i)} + b$ are even, forcing $v_K^{(i)}, v_I^{(i)} \ge 1$ — both A and B are sums of at least $k-1$ positive terms, so $\min(A,B) \ge k-1$, hence

$$\max(A,B) = \min(A,B) + |A-B| \ge (k-1) + |j|\,v_M.$$

The residue carrying this sub-class has modulus $2^{\max(A,B)+1}$, so it is visible at level L only when $L \ge \max(A,B)+1 \ge k + |j|\,v_M$, which is the claim. The bound is exercised against the enumerated sub-classes by `code/python/j_bound.py`. □

10. Asymptotic sharpness

Theorem 10.1 (Asymptotics of the density). *Let $q = 2^{v_M} - 1$. The refined bound of Corollary 8.3 gives the lower bound*

$$c_W^{(-q)} \ge \frac{1}{2q} + \frac{3}{2}\,\frac{2^{-2v_M}}{q}$$

(its further terms, of order $2^{-3v_M}/q$, being positive). Suppose moreover the completeness hypothesis *that, as $v_M \to \infty$, the universal families of Sections 5–7 exhaust the obstruction density up to order $2^{-3v_M}/q$:*

$$c_W^{(-q)} - (\textit{universal-family total}) = O\!\left(\frac{2^{-3v_M}}{q}\right),$$

equivalently, the aggregate contribution of the q-specific and level-(≥ 5) families is $O(2^{-3v_M}/q)$ (related to the sub-class growth of Conjecture 11.2). Then the matching upper bound holds and

$$c_W^{(-q)} = \frac{1}{2q} + \frac{3}{2}\,\frac{2^{-2v_M}}{q} + O\!\left(\frac{2^{-3v_M}}{q}\right) = \frac{1}{2q}\bigl(1 + O(2^{-v_M})\bigr),$$

so the simple lower bound $1/(2q)$ of Theorem 8.2 is asymptotically sharp. The hypothesis is corroborated by the density table (Table 1) but is not proved here; the sub-class growth it relies on is open (Conjecture 11.2).

Proof. The lower bound is Corollary 8.3: the principal class and the A-family sum exactly to $1/(2q)$ (Theorem 8.2), and the next-order families contribute its four correction terms — the B-family gives $2^{-2v_M}/q$, the C-family $2^{-2v_M-1}/q$, the F-family $2^{-3v_M}/q$, and G_1 gives 2^{-4v_M}. The two leading ones (the B- and C-families) sum to

$$\frac{2^{-2v_M}}{q} + \frac{2^{-2v_M-1}}{q} = \Big(1 + \tfrac{1}{2}\Big)\frac{2^{-2v_M}}{q} = \frac{3}{2}\,\frac{2^{-2v_M}}{q},$$

while the F-family $(2^{-3v_M}/q)$ and G_1 $(2^{-4v_M} = \Theta(2^{-3v_M}/q)$, since $q = 2^{v_M} - 1)$ are of the next order. For the matching upper bound, the order-$2^{-2v_M}/q$ mass is carried by the B- and C-families (their full towers), each level-4 q-specific family contributing only at the next order $2^{-4v_M} = O(2^{-3v_M}/q)$ and their number being bounded (Conjecture 11.1). Under the stated hypothesis the level-(≥ 5) and any remaining q-specific families add at most $O(2^{-3v_M}/q)$ in aggregate, so no further term arises at order $2^{-2v_M}/q$ and the upper bound matches the lower one; the relative correction to $1/(2q)$ is then $O(2^{-v_M})$. We stress that this aggregate-tail hypothesis is not established: the number of level-k sub-classes grows (Conjecture 11.2), and we have no a-priori bound forcing the tail below $O(2^{-3v_M}/q)$; the conclusion is therefore conditional. This is an asymptotic $(v_M \to \infty)$ statement, not a finite enumeration; the finite lower bound it sharpens, and the table values feeding the constant $\frac{3}{2}$, are checked by `code/python/atom_density.py`. □

11. Open problems

Conjecture 11.1 (No uniform q-specific form)**.** *At each v_M the level-k hierarchy contains finitely many q-specific sub-classes (e.g. four at $v_M = 4$, level 4) beyond the universal families, but there is no closed form for them uniform in v_M.*

Conjecture 11.2 (Sub-class growth)**.** *The number $\#\Sigma_k(v_M)$ of level-k sub-classes grows exponentially in k with ratio between 3 and 4; no second-order linear recurrence fits beyond three data points.*

Conjecture 11.3 (Spectral exponent)**.** *The growth exponent $\alpha(v_M)$ governing $|\mathcal{A}_L^{(-q,b)}|$, the spectral radius of the transfer matrix of the recursion, has no closed form; the empirical estimates remain L-dependent within the tested range. An operator-theoretic route in the spirit of the Wirsching operator for the $3n+1$ map [11] is the natural setting for this question.*

Conjecture 11.4 (Non-Mersenne density)**.** *For non-Mersenne q the principal class is absent (Corollary 3.3) and, empirically, $c_W^{(-q)} = \Theta\big(4^{-\mathrm{ord}_q(2)}\big)$, matching the generic positive-multiplier scale of [9] rather than the $\Theta(2^{-v_M})$ scale of the Mersenne ladder.*

Conjecture 11.5 (Sign conjugation)**.** *The positive-multiplier obstructions are not contained in the negative-multiplier ones, $\mathcal{O}^{(+q)} \not\subseteq \mathcal{O}^{(-q)}$; whether* *an arithmetic bijection (rather than the failed identity inclusion) relates the obstructions of the two signs is open.*

Funding and competing interests

The author received no funding for this research and declares no competing interests.

References

1. Daniel J. Bernstein and Jeffrey C. Lagarias, *The* $3x+1$ *conjugacy map*, Canadian Journal of Mathematics **48** (1996), no. 6, 1154–1169.
2. John H. Conway, *Unpredictable iterations*, Proceedings of the Number Theory Conference (Univ. Colorado, Boulder, 1972), University of Colorado, 1972, pp. 49–52.
3. Jeffrey C. Lagarias, *The* $3x+1$ *problem: an annotated bibliography (1963–1999) and (2000–2009)*, 2010, Two-part survey; Part II (2000–2009) is arXiv:math/0608208.
4. Jeffrey C. Lagarias (ed.), *The Ultimate Challenge: The* $3x+1$ *Problem*, American Mathematical Society, 2010.
5. K. R. Matthews and A. M. Watts, *A generalization of Hasse's generalization of the Syracuse algorithm*, Acta Arithmetica **43** (1984), no. 2, 167–175.
6. Herbert Möller, *Über Hasses Verallgemeinerung des Syracuse-Algorithmus (Kakutanis Problem)*, Acta Arithmetica **34** (1978), no. 3, 219–226.
7. Alain M. Robert, *A course in p-adic analysis*, Graduate Texts in Mathematics, vol. 198, Springer, 2000.
8. Jakob Stemberger, *Obstruction residues for the* $3n-1$ *map: full factor complexity, positive density, and a constructive lift*, 2026, Concept DOI (resolves to the latest version); source at https://github.com/yaccob/collatz.
9. ———, *Obstruction residues for the* $(an+b)$ *family: bias conjugation and multiplier order*, 2026, Companion manuscript, same repository (https://github.com/yaccob/collatz); archived under concept DOI 10.5281/zenodo.20356496.
10. Terence Tao, *Almost all orbits of the Collatz map attain almost bounded values*, Forum of Mathematics, Pi **10** (2022), e12, Preprint arXiv:1909.03562 (2019).
11. Günther J. Wirsching, *The dynamical system generated by the* $3n+1$ *function*, Lecture Notes in Mathematics, vol. 1681, Springer, 1998.

Independent researcher, Vienna, Austria
Email address: jakob.stemberger@yaccob.net
URL: https://orcid.org/0009-0001-2746-6877